

MACT: Multimodal Attenuation-Aware Pseudo-CT Synthesis for Whole-Body PET/MR

Marina Béguin¹[0009–0001–1464–5729]

ETH Zürich, Zürich, Switzerland mbegu@ethz.ch

Abstract. We present MACT, a multimodal framework for whole-body pseudo-CT synthesis for PET attenuation correction. MACT combines non-attenuation-corrected PET (NAC-PET), Dixon in-phase and out-of-phase MRI, and the X-ray topogram provided by the BIC-MAC challenge [1]. Complementary Dixon-derived contrast channels are explicitly constructed and adaptively reweighted using lightweight channel attention before dense prediction with a 3D U-Net. The training objective is designed around the downstream role of the predicted CT in PET reconstruction. In addition to CT-domain reconstruction and structural losses, the model is directly supervised in the 511-keV attenuation domain, globally and within attenuation-critical regions including the body contour, air/lung, bone, and dense bone. On the four public validation subjects, MACT achieved a whole-body SUV MAE of **0.035447**, organ bias of **2.221%**, CT μ -map MAE of **0.006035**, and brain outlier score of **0.040900**.

Keywords: Pseudo-CT synthesis · PET/MR · Attenuation correction

1 Introduction

Accurate attenuation correction is required for quantitative PET imaging. In PET/CT, attenuation maps can be derived directly from CT Hounsfield units (HU), whereas PET/MR requires attenuation information to be inferred from modalities that do not directly measure photon attenuation. The BIC-MAC challenge [1] provides several complementary inputs for this task. NAC-PET is acquired on the same PET/CT system as the reference CT and therefore provides a relatively reliable spatial reference, but contains limited direct information about dense structures such as cortical bone. Dixon MRI provides richer anatomical contrast but is acquired separately and is only approximately aligned with PET/CT space. Finally, the topogram is a two-dimensional X-ray projection acquired shortly before CT and contains complementary information about patient contour and skeletal anatomy.

Deep learning, and particularly convolutional architectures, has been successfully applied to MRI-based pseudo-CT synthesis for PET/MR attenuation correction [2, 3]. Building on this framework, we formulate pseudo-CT synthesis as an attention-based multimodal dense prediction problem. Since CT errors do

not contribute equally to PET attenuation correction, MACT is optimized not only for CT similarity but also directly for attenuation accuracy in anatomically and physically relevant regions.

2 Method

2.1 Multimodal representation

The model uses NAC-PET (P), Dixon in-phase MRI (M_{in}), Dixon out-of-phase MRI (M_{out}), and the topogram (T). MRI intensities are normalized independently for each subject using subject-wise robust intensity normalization. From these inputs, we construct an eight-channel multimodal representation by augmenting the original modalities with engineered Dixon interaction channels. Specifically, we derive:

$$M_{\text{mean}} = \frac{M_{\text{in}} + M_{\text{out}}}{2}, \quad (1)$$

$$M_{\text{diff}} = \frac{M_{\text{in}} - M_{\text{out}} + 1}{2}, \quad (2)$$

$$M_{\text{abs}} = |M_{\text{in}} - M_{\text{out}}|,$$

$$M_{\text{prod}} = M_{\text{in}} M_{\text{out}}. \quad (3)$$

The 2D topogram is resized to the corresponding spatial dimensions and replicated along its projection axis to form a 3D feature map. The resulting eight-channel representation is

$$X = [P, M_{\text{in}}, M_{\text{out}}, M_{\text{mean}}, M_{\text{diff}}, M_{\text{abs}}, M_{\text{prod}}, T]. \quad (4)$$

These engineered channels expose complementary relationships between the two Dixon acquisitions while retaining the spatially reliable PET and projection information.

2.2 Adaptive multimodal fusion

The eight input channels are passed through a squeeze-and-excitation-style channel-attention module [4] before spatial feature extraction. Global average pooling (GAP) summarizes each channel over the complete 3D patch, after which a two-layer projection generates channel-specific weights:

$$X' = X \odot \sigma(f_2[\phi(f_1(\text{GAP}(X)))]), \quad (5)$$

where ϕ denotes a LeakyReLU activation, σ denotes a sigmoid activation, and $\odot$ represents channel-wise multiplication.

This mechanism allows the relative contribution of PET, MRI-derived contrasts, and topogram information to vary according to the input rather than

treating all modalities equally. The reweighted representation is processed by a 3D U-Net architecture [5] with 24 base channels and a dropout probability of 0.05. The primary output is a continuous normalized pseudo-CT. During training, auxiliary heads additionally predict five CT-derived tissue categories corresponding to air/lung, fat, soft tissue, bone, and dense bone, together with a body mask. These auxiliary predictions regularize the shared representation but are not required during inference.

2.3 Attenuation-aware objective

A central design choice of MACT is to optimize the pseudo-CT according to its intended use for PET attenuation correction rather than relying exclusively on global HU error. The CT-domain component consists of a masked ℓ_1 regression loss in which bone and dense-bone voxels receive increased weights. Gradient and Laplacian terms encourage preservation of anatomical transitions and reduce excessive smoothing, while a tissue-calibration term promotes consistent HU prediction across broad tissue ranges. The model is additionally supervised directly in attenuation space. CT values are transformed to linear attenuation coefficients at 511 keV using a bilinear HU-to- μ conversion [6]. A global attenuation loss is complemented by regional attenuation losses targeting four regions that are particularly relevant for attenuation correction:

- the patient body contour,
- air and lung regions,
- bone,
- dense bone.

The complete objective can be summarized as

$$\mathcal{L} = \mathcal{L}_{CT} + \lambda_{\text{struct}} \mathcal{L}_{\text{struct}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\mu} \mathcal{L}_{\mu} + \sum_r \lambda_r \mathcal{L}_{\mu,r}. \quad (6)$$

For the submitted configuration, the global attenuation term receives a weight of 0.65. The regional μ -loss weights are 0.45 for the body contour, 0.30 for air/lung, 0.30 for bone, and 0.20 for dense bone. The CT reconstruction term remains the dominant image-domain loss, with additional weights of 0.06, 0.012, and 0.06 for gradient, Laplacian, and tissue-calibration losses, respectively.

3 Training and Experiments

The eight training subjects with full PET reconstruction data were retained as a diagnostic set and excluded from model training. The remaining labeled subjects were divided into 54 training and 13 validation subsets using an 80/20 split. Training was patch-based using 192^3 -voxel patches sampled from the labeled training volumes, with sampling constrained to informative patient/body regions.

We used AdamW with a learning rate of 2.2×10^{-4} , weight decay of 10^{-5} , batch size 1, cosine learning-rate annealing, and mixed-precision training. Training was performed on a single NVIDIA RTX 6000 Ada Generation GPU with 48 GB of VRAM using CUDA 12.6. Training was allowed to run for up to 1000 epochs, with early stopping based on validation CT μ -map MAE. At inference, overlapping 3D predictions with 50% overlap were combined into a full-resolution pseudo-CT. A patient-body mask estimated from the input features was then used to set voxels outside the body to air HU values before saving the final prediction.

Table 1. BIC-MAC evaluation results for the official baseline and the submitted MACT model.

Method	SUV MAE ↓	Organ Bias ↓	CT μ -MAE ↓	Brain Outlier ↓
BIC-MAC baseline [1]	0.0586	4.56%	0.00661	0.0541
MACT	0.0354	2.22%	0.00604	0.0409

MACT outperformed the official BIC-MAC baseline across all four reported metrics, with improvements in both CT attenuation accuracy and downstream PET reconstruction performance.

4 Discussion and Future Work

MACT achieved encouraging attenuation and PET reconstruction performance despite the limited amount of data available for model training. In particular, the eight BIC-MAC training subjects for which full PET reconstruction data were available were deliberately excluded from model optimization and retained as a diagnostic set. This allowed local PET reconstruction and analysis of how pseudo-CT errors propagated to downstream PET metrics, but further reduced an already small training cohort. In addition, the remaining subjects were divided using an 80/20 training-validation split. While this provided a sufficiently large validation set for model development, this split may be conservative given the limited cohort size.

Future experiments could investigate a smaller validation fraction or cross-validation to use the available data more efficiently. Once the model design and hyperparameters are fixed, the eight diagnostic subjects should also be reintegrated and the final model retrained using all available labeled BIC-MAC subjects.

A second limitation is that the current composite training objective was developed experimentally, and the relative contribution of its individual loss terms was not systematically optimized. A controlled ablation study is therefore needed to quantify the contribution of the CT-domain, structural, auxiliary, global attenuation, and regional attenuation losses. Systematic optimization of their relative weights could further improve the trade-off between CT μ -map accuracy and the

downstream PET-based challenge metrics.

The limited dataset size also motivates the use of external pretraining. Paired MRI-CT datasets such as SynthRAD [7] could be used to pretrain the encoder and auxiliary tissue or material prediction heads before multimodal fine-tuning on BIC-MAC. Rather than transferring a complete MRI-to-CT synthesis model, pretraining could focus on learning transferable representations of attenuation-relevant tissue classes such as air, fat, soft tissue, bone, and dense bone, thereby reducing sensitivity to differences in MRI acquisition protocols.

Finally, future work could investigate region-specific refinement, particularly around the skull, air cavities, and other head structures, to improve local attenuation accuracy while preserving the whole-body performance of the current model.

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